
3D Pose Estimation using mmWave Radar Data

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Abstract

This report investigates 3D pose estimation from sparse mmWave radar point cloud data using the mini-MMFi dataset. We evaluate four deep learning approaches using different feature templates to address the challenge of data sparsity: MFPC with 3D CNNs, MFPC with 2D projection maps and 2D CNNs, temporal sequence models using LSTMs, and point based networks. Using a cross-subject evaluation protocol with 10 subjects performing 27 actions, we find that MFPC combined with 3D CNN achieves the best performance, while the 2D projection approach offers comparable accuracy with significantly faster training. Upper body joints demonstrate robust estimation, while lower body joints, particularly the pelvis and ankles, show substantially higher errors due to environmental noise. These results demonstrate the efficacy of deep learning with feature template representations at estimating pose from sparse mmWave point clouds, but that a significant limitation remains with environmental reflections.

1 Introduction

Radio-frequency based sensing uses electromagnetic waves to sense the environment without using cameras, which preserves privacy for users and often doesn't require additional hardware, such as using the existing WiFi router in people's homes.

WiFi sensing uses Received Signal Strength Information (RSSI) or Channel State Information (CSI) to identify patterns, which can be effective for basic applications like localisation. In contrast, mmWave sensing uses radar that operates in higher frequency bands, allowing for significantly higher resolution and the generation of detailed point cloud representations of the environment. This allows mmWave systems to perform tasks that require more accurate information, such as the 3D human pose estimation we're addressing in this report.

At the core of mmWave radar technology is the FMCW signal, referred to as a chirp, which is transmitted and reflected back from targets with a delay related to distance. By mixing the transmitted and received signals and analysing them across time and antennas, the radar estimates range, velocity, and angle of arrival, enabling reconstruction of a 3D point cloud of the scene. These sensing capabilities enable a range of privacy preserving human sensing applications, from health monitoring to gesture analysis. In this report, we focus on 3D pose estimation, which aims to estimate a subject's joint structure from sparse mmWave point cloud data.

The dataset being used, mini-MMFi, reflects a set of ten human subjects performing 27 different actions. For each action they perform, we have the mmWave data in the form of a point cloud, and the ground truth joint positions which have been calculated from images of the subject in the same timeframe.

Figure 1 shows an example of the mmWave point cloud data for a single frame and the ground truth joint positions for a specific frame in the dataset. We can see from 1a that the point cloud is very sparse, containing only 7 radar reflection points, which is fewer than even the 17 joint positions

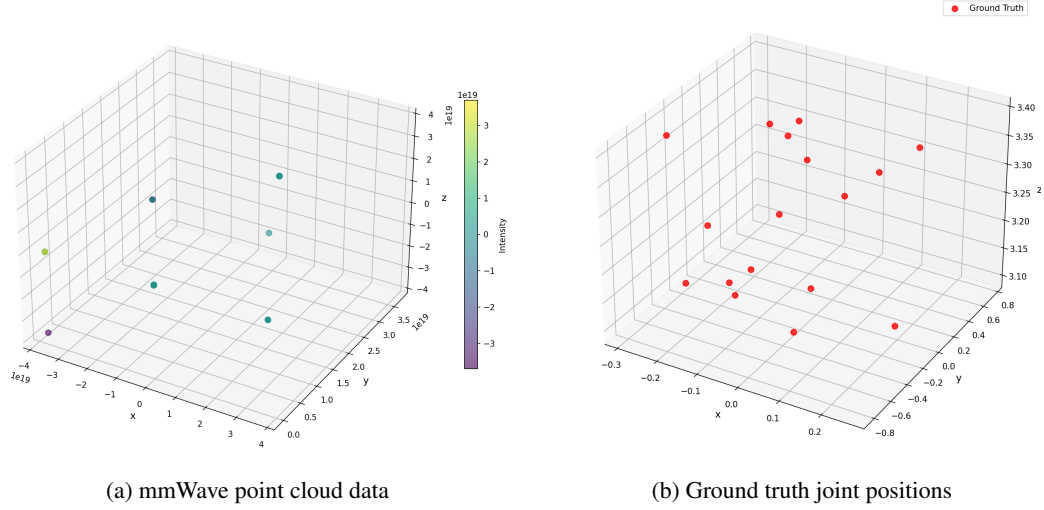


Figure 1: Visualisation from the dataset (Subject 01, Action 05, Frame 10)

present in the ground truth 1b. The points are coloured by their intensity values, representing the strength of the radar reflection at each location, which provides an additional feature beyond the coordinates that may help the model distinguish different poses. The disparity between the number of reflected points and the number of joints highlights a key challenge of the pose estimation task - the model needs to infer the 3D pose from data that is sparse, noisy, and unstructured.

2 Methodology

2.1 Method 1: MFPC with Voxelisation and 3D CNN

Multi-Frame Point Cloud (MFPC) Representation Using a MFPC representation is one way of addressing the challenges of sparsity and lack of structure in the data. Rather than using only the frame at time t , MFPC forms a stacked cloud by merging the current frame with the previous $k - 1$ frames. This increases the density of the point cloud, as points that may be missing in a single frame can appear in neighbouring frames and collectively provide more complete information about the body, from which the pose can be inferred. This process is visualised in Figure 2. There is a trade-off with the value of k we choose to use; larger windows increase density and robustness, but can reduce the signal when body parts move quickly, as points will appear at different positions across different frames.

Voxel Grid Representation To convert a 3D point cloud into a fixed size tensor, we can use voxelisation [4]. This divides the 3D space into a grid of cubes, called voxels, with the value of each entry defined by the presence of a point from the point cloud being within it. Once converted to this representation, the input tensor resembles a 3D image and can be used by a CNN. The drawback is that voxelisation is compute and memory expensive, since the cost scales with the volume of the grid rather than the number of points. There is also a resolution trade-off, where larger grids preserve more detail but increase sparsity and memory usage, while smaller grids are cheaper but can lose information where points are close together.

Convolutional Neural Network (CNN) for 3D joint regression A CNN can be used to learn the relationship between the points and the joints in the pose. The network progressively builds higher level features through stacks of 3D convolutions with non-linearities (ReLU) and pooling, providing robustness to spatial perturbations. A linear layer then maps the extracted features to joint coordinates. In our approach, we combine the MFPC and voxel grid feature templates with a 3D CNN 2.

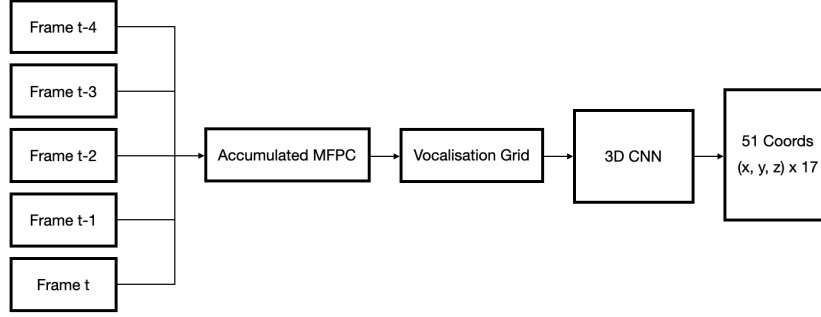


Figure 2: MFPC pipeline

2.2 Method 2: MFPC with 2D Projection Maps and 2D CNN

An alternative to voxelisation is to project the 3D point cloud into a set of 2D projection maps, which function like heatmaps where the value of each pixel is defined by aggregating the points that fall in that position. In this representation, points are accumulated into three tensors, representing the front view, top view, and side view, shown in Figure 3. In our implementation we set a pixel to 1 if any point falls within it, otherwise 0. These maps can then be treated as a multi-channel 2D input and processed by 2D CNNs, like ResNet [3], which are faster than 3D convolutions. The drawback is the projection loses depth information which could be useful for the pose estimation task, though this is mostly offset by using multiple views.

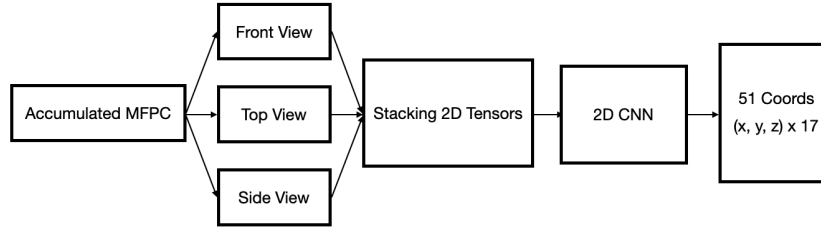


Figure 3: 2D projection map pipeline

2.3 Method 3: Temporal Sequence Models

Rather than combining multiple frames into a single representation, temporal sequence models process the data over time (frame by frame) and learn how the pose changes to be able to estimate it more effectively. Recurrent architectures, such as LSTMs, maintain a hidden state as a form of memory of past observations, allowing the model to use motion context when predicting the current pose. In our implementation, each frame is converted into 2D projection maps and encoded into a feature vector using a shared 2D CNN; the resulting sequence of feature vectors is then passed through an LSTM, with a final linear layer predicting the joint coordinates. This is visualised in Figure 4.

2.4 Method 4: Point Set Processing and Point Based Networks

Point set processing methods aim to operate directly on the raw mmWave point cloud, rather than converting it to a fixed size tensor, as this can be inefficient when the point cloud is sparse, and it avoids the resolution to accuracy trade-off mentioned in the voxelisation method. Point based networks such as PointNet [2] use this approach by applying the same MLP to every point to project it into a larger feature space. Max pooling is then used on these outputs to produce a single feature vector that is invariant to the order of points, which is then run through a linear layer to predict the joint coordinates. In our implementation, the input to the network is a set of points with shape $(N, 5)$, where each point is (x, y, z, d, I) corresponding to position, Doppler, and intensity.

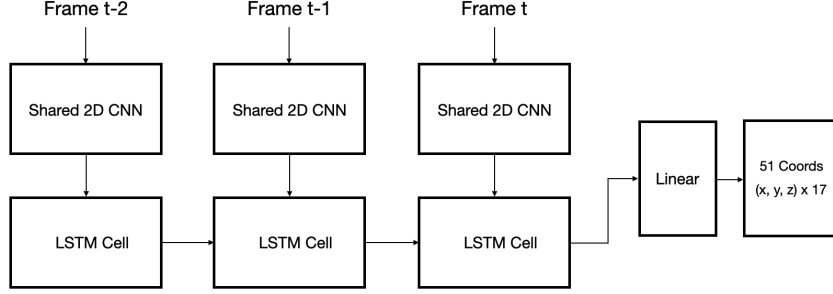


Figure 4: Temporal LSTM architecture

3 Experiments

We evaluate on the mini-MMFi dataset [1], a subset of the MM-Fi multimodal wireless sensing dataset captured in a single indoor environment. It contains data from 10 subjects, each performing 27 actions. Each frame consists of a variable length mmWave radar point cloud where each point is (x, y, z, d, I) (3D location, Doppler velocity, and signal intensity), and a ground truth 3D pose.

For performing our experiments, we use a cross-subject experimental protocol, where we use the first 8 subjects for training, subject 9 for validation, and subject 10 for testing, to measure generalisation to unseen subjects and avoid data leakage that could be subject specific. We use a mean absolute error (MAE) and root mean squared error (RMSE) to evaluate the performance of each model on the test and validation sets. In our data processing pipeline, each frame is first filtered to remove points with low Doppler and low intensity, which are likely to be noise and contain little signal. All the models are trained using the mean squared error (MSE) loss between predicted and ground truth joint locations. We also use batch normalisation and dropout to improve convergence and prevent overfitting.

We set the standard model hyperparameters for each model depending on the convergence observed in the validation loss, which resulted the parameters in Table 1.

Hyperparameter	MFPC + 3D-CNN	MFPC + 2D-CNN	Point Based Network	Temporal + LSTM
Learning Rate	3×10^{-5}	1×10^{-4}	1×10^{-4}	1×10^{-4}
Batch Size	512	512	512	512
Epochs	10	10	10	10

Table 1: Hyperparameters for each model

Figure 5 shows a successful prediction from the Point Based Network, showing the model’s predictions closely matching the ground truth.

Conversely, Figure 6 demonstrates a failure case where the model struggles to accurately estimate the pose. This error is particularly visible in the top-down view, where the predicted pose shows a clear skew away from the actual subject position. Errors like this could be caused by motion of the subject or occlusion, making the signal in the data sparse and difficult for the model to interpret.

Table 2 presents the results of the models on the test set. The MFPC + 3D-CNN model with a window size of $k = 10$ achieves the lowest errors (MAE 0.0863), slightly outperforming the $k = 5$ configuration (MAE 0.0888). This improvement indicates that a larger window size helps to mitigate the sparsity of the radar data, and this is a slightly better balance of the trade-off discussed in the methodology. The MFPC + 2D-CNN also performs strongly (MAE 0.0866), matching the best 3D-CNN model, suggesting the 2D projection maps do not lose too much information, and this model was also significantly faster to train. The Temporal + LSTM model and Point Based Network show slightly higher errors, indicating that the convolutional architectures on accumulated frames may be marginally more robust to the sparsity and noise in the mmWave data.

The joint-wise error analysis in Table 3 shows upper body joints (head, neck, shoulders) achieve the lowest errors, while lower body joints show significantly higher errors, particularly the pelvis and left

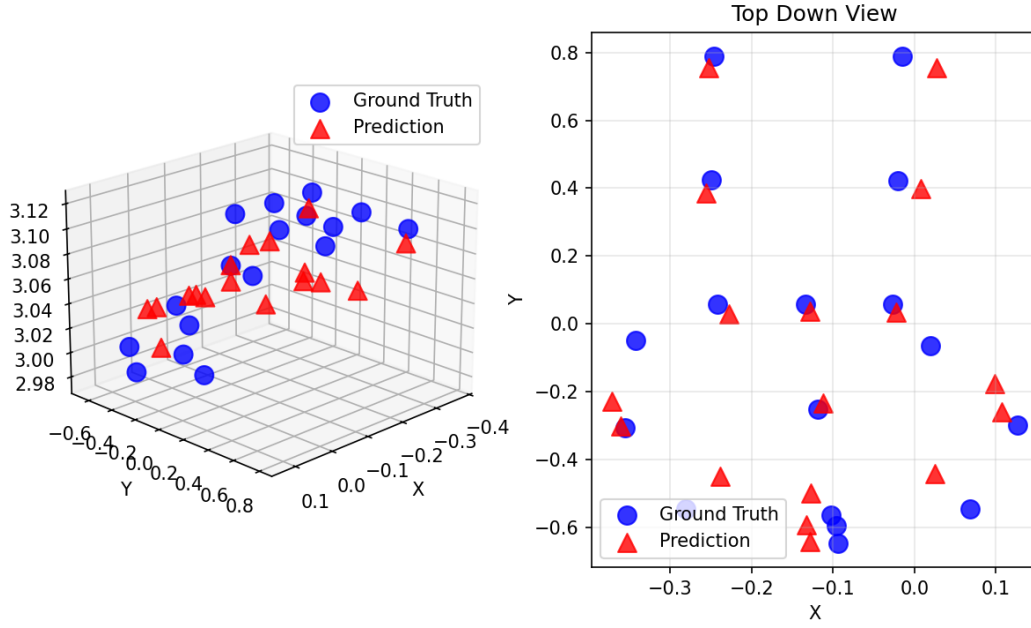


Figure 5: Successful prediction from the point based model

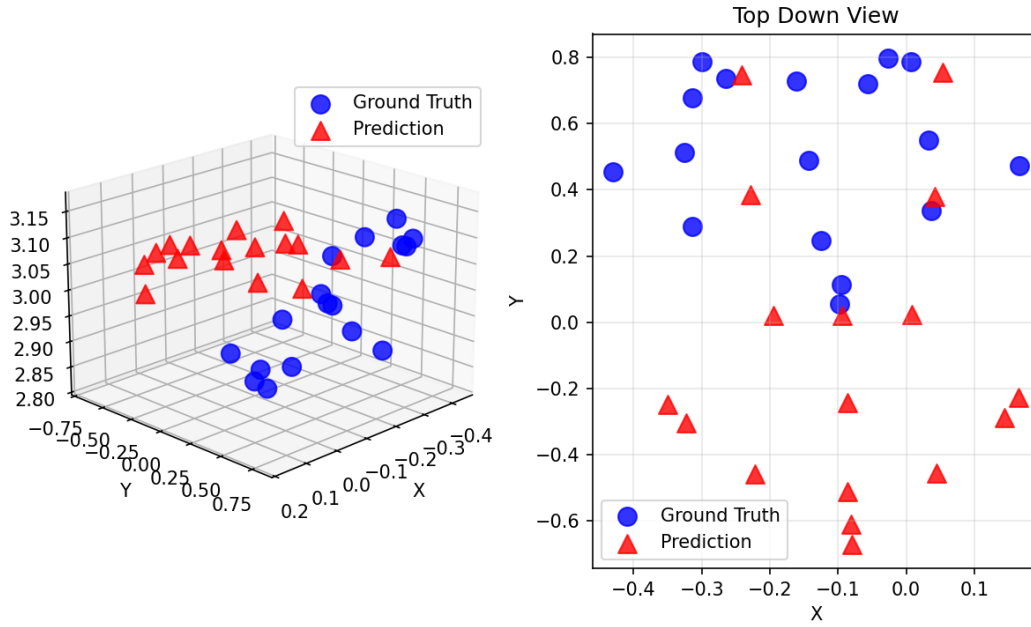


Figure 6: Failure case from the LSTM model

127 ankle. This could be caused by reflections from the floor or other objects on the ground leading to
 128 more noise in the data, making it difficult to distinguish the exact position of lower body joints. The
 129 3D CNN and 2D Projection models consistently outperform the LSTM and point based approaches
 130 across nearly all joints, with the 2D Projection method showing particularly strong performance on
 131 the neck and hip joints. However, the point based model achieves the best performance for the left
 132 ankle, suggesting point based representations may better capture the motion of lower body joints.

Model	Experimental parameters	MAE*	RMSE*
MFPC + 3D-CNN	$k = 5$	0.0888	0.1331
MFPC + 3D-CNN	$k = 10$	0.0863	0.1322
MFPC + 2D-CNN	–	0.0866	0.1321
Temporal + LSTM	seq_len = 5	0.0894	0.1336
Point Based Network	–	0.0923	0.1358

Table 2: Model results on the test set

* Averaged over 3 training runs

Joint	MAE				RMSE			
	3D CNN	2D Proj	LSTM	Point Based	3D CNN	2D Proj	LSTM	Point Based
Head	0.132	0.132	0.134	0.149	0.161	0.161	0.161	0.177
Neck	0.138	0.132	0.134	0.146	0.165	0.160	0.162	0.174
RShoulder	0.138	0.134	0.140	0.155	0.168	0.168	0.169	0.184
RElbow	0.129	0.130	0.129	0.144	0.163	0.164	0.164	0.178
RWrist	0.139	0.138	0.139	0.147	0.168	0.167	0.166	0.176
LShoulder	0.147	0.142	0.141	0.151	0.174	0.172	0.169	0.179
LElbow	0.148	0.145	0.145	0.155	0.172	0.172	0.169	0.182
LWrist	0.143	0.135	0.145	0.157	0.171	0.167	0.171	0.184
RHip	0.165	0.154	0.166	0.182	0.193	0.189	0.194	0.208
RKnee	0.194	0.190	0.203	0.215	0.220	0.217	0.226	0.238
RAnkle	0.194	0.188	0.201	0.212	0.219	0.216	0.224	0.235
LHip	0.158	0.152	0.160	0.177	0.191	0.190	0.192	0.207
LKnee	0.221	0.220	0.225	0.229	0.268	0.267	0.270	0.272
LAnkle	0.365	0.365	0.369	0.361	0.410	0.408	0.412	0.404
Chest	0.156	0.149	0.163	0.177	0.188	0.186	0.192	0.206
Spine	0.214	0.209	0.217	0.229	0.261	0.257	0.262	0.273
Pelvis	0.358	0.357	0.363	0.367	0.406	0.404	0.408	0.412

Table 3: Joint-wise MAE and RMSE errors across all models

4 Conclusion

In this report, we evaluated four deep learning architectures for 3D pose estimation using mmWave radar data from the mini-MMFi dataset. Our experiments demonstrate that addressing the sparsity of radar point clouds is critical for accurate pose estimation, though multiple feature templates and model architectures can be used to achieve this. The MFPC approach combined with a 3D CNN achieved the best overall performance by accumulating the point cloud over multiple frames. MFPC with 2D Projection maps offered a strong trade-off, nearly matching the accuracy of 3D convolutions while being significantly faster to train. However, we identified clear limitations regarding lower body detection. While upper body estimation was robust, joints such as the ankles had high error rates, likely due to noise from floor reflections. Future work should focus on advanced noise filtering to mitigate environmental reflections reducing the performance of estimating lower limb positions, and investigate architectures like Transformers, which can provide strong computational efficiency while leveraging greater context awareness to improve accuracy.

References

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